Kuta Software - Infinite Calculus

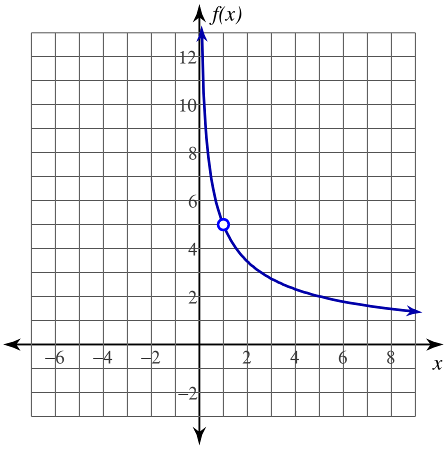
Name________________________________

# L'Hôpital's Rule

Date________________ Period____

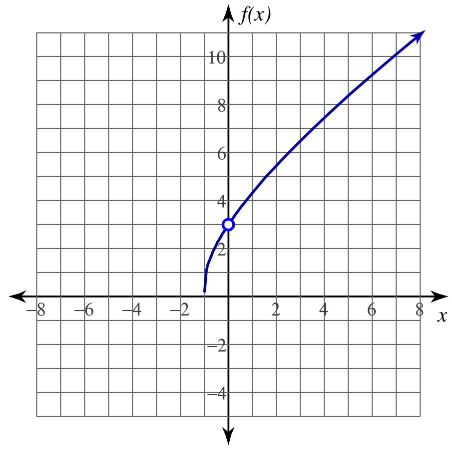
**Evaluate each limit using L'Hôpital's Rule.**

1) $\lim_{x \to 1} \frac{5 \ln x}{x - 1}$




5

2) $\lim_{x \to 0} \frac{3x}{\ln (x + 1)}$




3

3) $\lim_{x \to 0^+} 5x^2 \ln x$

0

4) $\lim_{x \to \infty} 4x \cdot e^{-x}$

0

5) $\lim_{x \to \frac{\pi}{2}} (3\sec x - 3\tan x)$

0

6) $\lim_{x \to \infty} \left( \frac{x^2}{x - 1} - \frac{x^2}{x + 1} \right)$

2

7) $\lim_{x \to 0^+} 5 \cdot (\tan x)^{\sin x}$

5

8) $\lim_{x \to 0^+} 3x^x$

3

**Evaluate each limit. Use L'Hôpital's Rule if it can be applied. If it cannot be applied, evaluate using another method and write a * next to your answer.**

9) $\lim_{x \to 0} \frac{e^x - e^{-x}}{x}$

2

10) $\lim_{x \to 0^+} \frac{e^x + e^{-x}}{\sin (2x)}$

$\infty$ *